

Post-quantum migration for agentic AI systems in financial services to secure agent identities and communications.

Finance, Banking and Insurance × Post-quantum cryptography migration × Agentic AI · OS083 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
75 is the world moving	63 can we play, can we win	HIGH 1 named in the evidence	€25.4m modelled evidence · per year	LATER research and standards activi...	L1 14 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

Evidence gap (SC-13). Orange has 4.0 published reference(s) in Finance, Banking and Insurance, below the threshold of 5. A customer conversation here has no proof point behind it yet.

The opportunity

This opportunity helps financial services firms secure their agentic AI systems against future quantum threats by migrating to post-quantum cryptography. It targets CISOs and CTOs in banks and insurers who are deploying AI agents and need to protect agent identities and communications now, before quantum computers become a reality.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€31.2m €5.6m – €150m	€25.4m €4.6m – €122m	€711k €128k – €3.4m	modelled
Observed: tendered contract value, annualised	€100m	€100m	€2.8m	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Finance, Banking and Insurance (EU27_2020)	11,116 enterprises	observed	Eurostat · sbs_sc_ovw	2024
Enterprises adopting Agentic AI	5.3 % of enterprises (10+ employees)	proxy	Eurostat · isoc_eb_ain2 · E_AI_TPA	2025
Annual value of one engagement	€240k per enterprise per year	assumption	config/sizing.yaml · fallback_bands_eur	size-2026-08-a
Size-weighted buyer base	2,426 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	2.8 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Adoption rate is a declared proxy. AI automating workflows or assisting decisions Part of this vertical is not covered separately by the enterprise ICT survey, so the all-activities rate was used for those slices. Contract value is a configured assumption, not observed. Only 2 matching tender notices, below the 6 needed for a robust median, so the configured band for this business domain is used. It is an assumption owned by innovation-radar-curator, not evidence. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders. 1 of 2 declared geographies are outside the European reference data (US); the estimate covers the European footprint only. Sector adoption is read from Eurostat's all-activities aggregate for part of this vertical, which the enterprise ICT survey does not cover separately: PROXY — ICT survey excludes the financial sector; PROXY — auxiliary financial activities; PROXY — insurance and pension funding

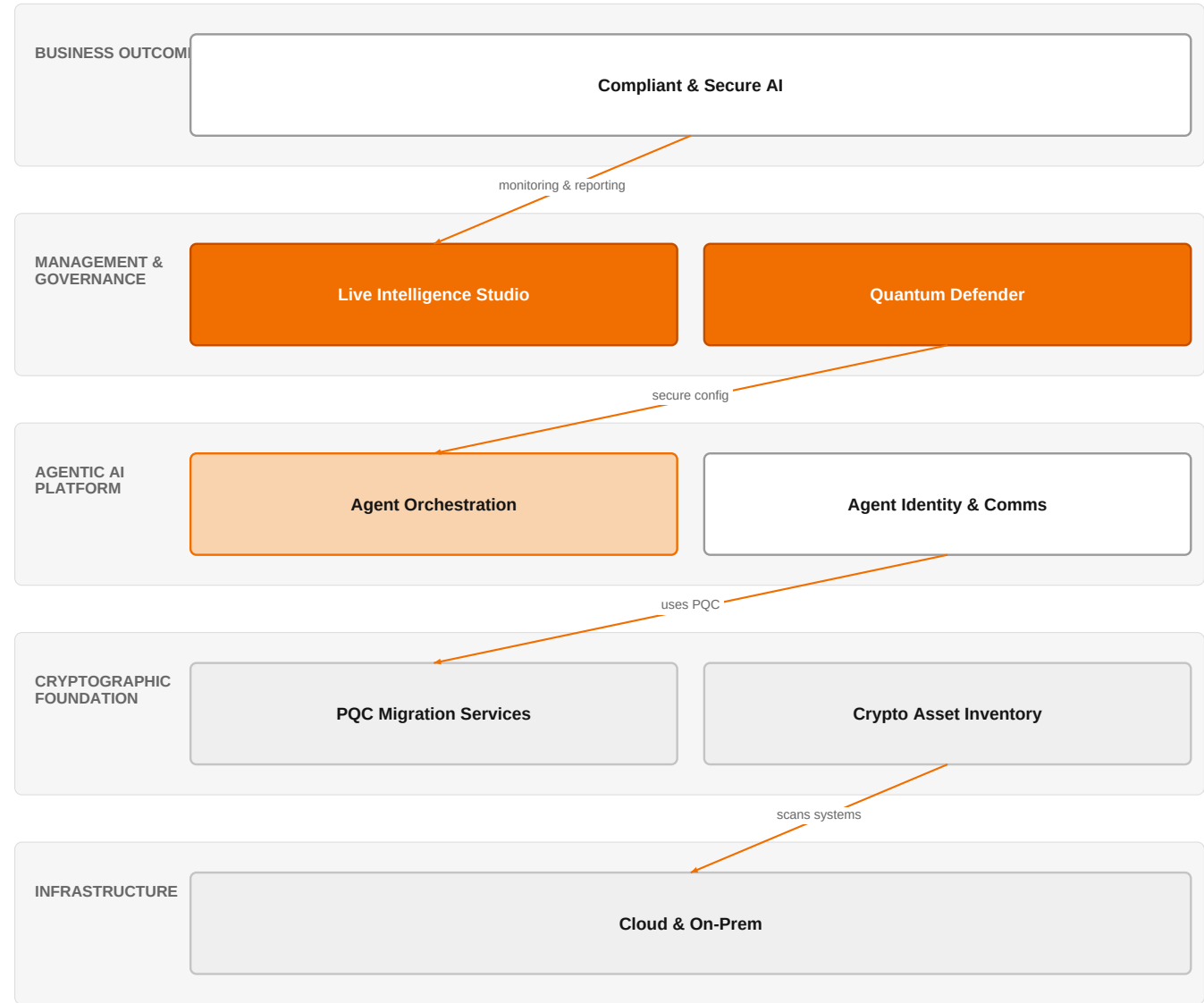
Observed: tendered contract value, annualised

Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€100m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-26..2026-08-05
Assumed obtainable share of the serviceable market	2.8 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 8 matching notices, matched on vertical via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size. 1 of 2 declared geographies are outside the European reference data (US); the estimate covers the European footprint only.

The solution, in one picture

Post-Quantum Security for Agentic AI



This diagram shows how Orange's quantum-safe services and management tools secure the agentic AI stack, from infrastructure up to business outcomes.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

Agentic AI systems increasingly rely on classical public-key cryptography for agent identity, tool invocation, inter-agent communication, model integrity, and persistent state, exposing them to quantum threats along confidentiality and integrity axes. Adversaries can already record encrypted traffic to decrypt later, and the migration is a complex, multi-year undertaking requiring larger parameter sizes, hybrid schemes, and ecosystem coordination. Regulated enterprises face pressure from second-line-of-defense functions to make migration auditable and aligned with regulator expectations.

Sources: SIG-5280658BD549 Computers 2026-07-08; SIG-81E56EEBF9D4 openalex.org 2026-06-15; SIG-5C7568EBD975 Elsevier BV 2026-01-01; SIG-8BD0B85D3181 openalex.org 2025-12-24; SIG-ED58CE7F1766 openalex.org 2025-11-24

Who buys, and why now

The Chief Information Security Officer (CISO) signs, but the Chief Technology Officer (CTO) and AI platform owners feel the pain because their agentic AI deployments depend on classical cryptography that will break. The trigger is often a

regulatory audit or a risk assessment by the second line of defense that flags the lack of a post-quantum migration plan. Operational pain includes inability to meet compliance deadlines, risk of harvest-now-decrypt-later attacks on sensitive financial data, and the complexity of retrofitting quantum-safe measures into live agent systems.

Why it is hot now — every claim bound to a dated source

- Agentic AI systems rely on classical public-key cryptography, exposing them to quantum computer threats.

[SIG-5280658BD549, SIG-81E56EEBF9D4]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a team of Cybersecurity experts and AI/Data/Cloud experts to conduct a cryptographic asset discovery and inventory of the customer's agentic AI systems, using Orange Group researchers for methodology. Then, leveraging the Quantum Defender / quantum-safe offers, Orange would design and implement a migration roadmap, integrating with Microsoft and Genesys platforms where relevant. The engagement would follow ISO 27001 and SOC 2 Type II processes, and could be delivered through Live Intelligence Studio for monitoring and management.

Why Orange can win

Orange brings a unique combination of cybersecurity expertise, quantum-safe offers, and deep experience in regulated industries, backed by Orange Group researchers. The ability to provide a sovereign, certified (ISO 27001, SOC 2 Type II, CISPE) solution is a differentiator. However, the assets are thin: there is no specific post-quantum migration service offering beyond Quantum Defender, and no named reference in financial services for this exact use case.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Live Intelligence Studio	offer	L1	offer provides the technology in a matching domain	unconfirmed
Quantum Defender / quantum-safe offers	offer	L1	offer addresses the use case but not this technology	unconfirmed
Microsoft	partner	L2	partner provides the required technology	unconfirmed
Genesys	partner	L2	partner provides the required technology	unconfirmed
ISO 27001	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
SOC 2 Type II	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
CISPE	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
AI / Data / Cloud experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Cybersecurity experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Digital experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Defense and security experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Healthcare experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Banqsoft	reference	SUP	published reference in this vertical, different use case	unconfirmed
Stø	reference	SUP	published reference in this vertical, different use case	unconfirmed

Competitive landscape

HIGH competitive intensity (score 8.9; 1 of 8 listed players appear in this topic's own evidence)

Crowded, with heavyweight incumbents already visible in the evidence. Win on a specific differentiator or do not bid.

Microsoft, as a hyperscaler and Orange partner, sells agentic AI and has strong cloud capabilities, but its quantum-safe offerings are less specialized. Accenture, Capgemini, and Infosys are global system integrators active in finance and banking, with broad consulting and implementation strength. Genesys, NiCE, and Salesforce focus on customer experience platforms and may embed agentic AI but lack deep security expertise. Mistral AI provides AI models but not end-to-end security migration. The customer will compare Orange's security pedigree and sovereign credentials against the scale and AI-native capabilities of these competitors.

Sources: SIG-5C7568EBD975 Elsevier BV 2026-01-01

Competitor	Category	Position here	Basis	Evidence
Microsoft	Hyperscaler	sells Agentic AI; competes in Cloud — also an Orange partner	evidenced	SIG-5C7568EBD975
Accenture	Global systems integrator	sells Agentic AI; active in Finance, Banking and Insurance; competes in Cloud	structural	—
Capgemini	Global systems integrator	sells Agentic AI; active in Finance, Banking and Insurance; competes in Cloud	structural	—
Infosys	Global systems integrator	sells Agentic AI; competes in Cloud	structural	—
Genesys	Customer-experience platform	sells Agentic AI — also an Orange partner	structural	—
NiCE	Customer-experience platform	sells Agentic AI — also an Orange partner	structural	—
Salesforce	Customer-experience platform	sells Agentic AI	structural	—
Mistral AI	Data and AI platform	sells Agentic AI; competes in Cloud	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 What agentic AI systems are you currently deploying or planning, and what cryptographic mechanisms do they use for agent identity and inter-agent communication?
- 2 Have you conducted a cryptographic asset inventory of your AI and non-AI systems? If so, what coverage does it have for agentic AI components?
- 3 What is your current timeline for post-quantum migration, and have you received any regulatory or audit pressure to start?
- 4 Who in your organization owns the security of AI agents—CISO, CTO, or a dedicated AI governance team?
- 5 Are you using any of the major agentic AI platforms (e.g., Microsoft, Genesys) that might have their own quantum-safe roadmaps?
- 6 What is your biggest concern: harvest-now-decrypt-later data exposure, or integrity of agent identities and tool invocations?

Objections you will meet

They will say	You can answer
Quantum computers are years away; why should we invest now?	It's true that a cryptographically relevant quantum computer may be years away, but adversaries are already recording encrypted traffic to decrypt later. For financial services, the data has a long shelf life, and migration is a multi-year effort. Starting now reduces the window of exposure.
Our agentic AI platforms (e.g., Microsoft, Genesys) will handle security themselves.	Platforms provide some security, but the overall architecture—including custom agents, tool integrations, and persistent state—is your responsibility. Our assessment can identify gaps that platform vendors don't cover.
We already have a post-quantum migration plan for our core systems.	That's a good start, but most plans focus on static, operator-controlled systems. Agentic AI introduces dynamic, autonomous components that require different migration strategies. We can help extend your plan to cover these.

Risks and unknowns

The main risk is that the customer may not yet perceive quantum threats as urgent, especially for agentic AI which is still emerging. The migration is complex and multi-year, and there is no standardized approach for agentic AI specifically. Unknowns include the timeline for a cryptographically relevant quantum computer, the evolution of NIST standards, and whether the customer's agentic AI platforms (e.g., Microsoft, Genesys) will support post-quantum algorithms in time.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief for a financial services client on post-quantum migration, leveraging Orange's Quantum Defender and Live Intelligence Studio, and include references to ISO 27001 and SOC 2 Type II certifications to differentiate the proposal.
Sales	In a conversation with a financial services CIO, say: 'As agentic AI becomes more prevalent, the security of agent identities and communications is paramount. Orange's Quantum Defender offers a path to quantum-safe operations, and we can help you assess your readiness.'
Strategist / Innovator	Commission a deep-dive brief on post-quantum migration for agentic AI in financial services, focusing on how Orange's Quantum Defender and Live Intelligence Studio can be positioned, and identify which standards bodies and research initiatives to monitor.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-40B64E273856	2026-07-30	International Journal of ...	1	Post-Quantum Cryptography Migration for Enterprise Security
SIG-B8561992126B	2026-07-30	openalex.org	1	Post-Quantum Cryptography Migration for Enterprise Security
SIG-5280658BD549	2026-07-08	Computers	1	Post-Quantum Cryptography Migration for Agentic AI Systems
SIG-81E56EEBF9D4	2026-06-15	openalex.org	1	Post-Quantum Cryptography Migration for Agentic AI Systems
SIG-969C5FDB15DD	2026-02-02	Institute of Electrical a...	1	Post-Quantum Cryptography Migration Framework for Real-Time Payment Gateways
SIG-13AAE2CD702E	2026-01-19	MDPI AG	1	Cryptographic Asset Discovery and Inventory for Embedded Systems: A Framework for Post-Quantum ...
SIG-6CA0BD99D256	2026-01-08	2026 Second International...	1	Post-Quantum Cryptography Migration Framework for Real-Time Payment Gateways
SIG-5C7568EBD975	2026-01-01	Elsevier BV	1	The Role of the Second Line of Defense in Post-Quantum Cryptography Migration and Cryptographic...
SIG-D81DF125FECB	2026-01-01	Proceedings of the 23rd I...	1	A Capability-Driven Framework for Prioritizing Post-Quantum Cryptography Migration in Enterpris...
SIG-8BD0B85D3181	2025-12-24	openalex.org	1	Enterprise Migration to Post-Quantum Cryptography: Timeline Analysis and Strategic Frameworks
SIG-ED58CE7F1766	2025-11-24	openalex.org	1	Estimating Migration Timelines for Enterprises Transitioning to Post-Quantum Cryptography: A Re...
SIG-B6E57CA5BED7	2026-08-17	americanbanker.com	2	Beyond the bot : The agentic AI shift in Financial Services
SIG-C0EC2AA3673A	2026-08-12	TradingView	2	AI agents for insurers: IT service provider adesso acquires leading AI claims platform omni:us ...
SIG-15F7DBC050C1	2026-07-28	technologyreview.co m	3	Building the enterprise environment for agentic AI

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: diagram (2 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:14+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:17+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-19T04:56:44+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.